

Convolutional Neural Network Modeling for Pest Detection in Corn Crops: Optimization for Monitoring Efficiency

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Abstract—This project aims to develop automated techniques for early detection of pests in maize crops using computer vision and deep learning. Early detection of pests is key to crop success, but is typically done manually, which is time consuming and resource intensive. Accurate methods are needed to obtain data quickly and non-invasively. The implemented method consists of analysing images of maize crops using convolutional neural networks(CNN) trained to identify insects. The results show that the developed approach achieves an accuracy of over 95% in detecting the main pests studied, surpassing the accuracy of manual methods. The data generated by computer vision and deep learning techniques can be useful to farmers in making decisions about preventive pest control.

Index Terms—Pest detection, Fall Army-worm, Maize Crops, Computer vision, Deep learning, Predictive control.

I. INTRODUCTION

Detecting pests in maize crops has always been a challenge, as it can significantly impact both yield and quality. However, technological advancements in detection and monitoring have provided new opportunities to address this issue. The purpose of this study is to explore innovative solutions to improve maize welfare and yield, which is crucial for food security and economies [1]. Early and accurate detection is essential for effective control. Computer vision and deep learning have been shown to be effective in detecting diseases in crop images, allowing for accurate assessments [2].

Traditionally, the visual diagnosis of plant diseases has limitations as it requires experience and access to experts [3]. This technology focuses on classifying and identifying images to accelerate the detection of diseases and pests in maize. It achieves high accuracy and reduces costs.

The proposed methodology follows an incremental approach, gradually developing the pest detection model. To achieve this, we will perform iterations by improving training parameters and image pre-processing. We will evaluate the accuracy and performance of the model using metrics such as precision and MAP. The aim of this project is to achieve a reliable and accurate detection of pests in maize. Our goal is to achieve an accuracy of at least 90%.

Chapter II addresses the problem of pest detection in maize crops and includes a review of the current state of the art. The research objectives are outlined in Chapter III, and Chapter IV describes the methodology for developing the pest detection model. Chapter V presents the preliminary results, followed by final conclusions in Chapter VI. Chapter VII is dedicated to expressing gratitude to the collaborating institutions.

II. CONTEXT

Computer vision and image recognition have significantly advanced the detection of diseases and pests in maize. These technologies are now indispensable in agriculture, providing effective and accurate solutions to identify and manage problems that affect maize productivity and quality. Computer vision has developed rapidly and has been successfully applied in various industries [1]. In agriculture, the use of this technology to detect disease in maize is crucial. The ability to achieve high accuracy and reduce costs has revolutionised the approach to crop diseases and pests [4].

CNN will provide valuable information to farmers for the detection and monitoring of pests. The use of open systems allows for the discussion of smart agriculture, which integrates multiple technologies to optimize certain processes [5]. Artificial intelligence will also play a crucial role in analysing and processing large amounts of collected data, enabling more accurate and faster decision-making for crop disease management.

A. Background Research

The use of CNN for pest detection in maize crops has gained considerable attention and has been recommended by many authors in the field of precision agriculture and artificial intelligence applied to agriculture. Figure 1 shows a timeline of emerging technologies in agriculture.

Roboflow and YOLO V8 were employed in this study's framework to perform object detection with an emphasis on identifying pests in maize fields. Roboflow demonstrated remarkable performance as evidenced by its mAP of 98.1%, accuracy of 97.5%, and recall of 96.8% following 300 training

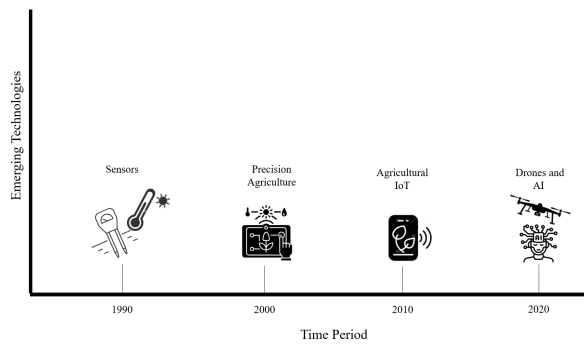


Fig. 1. Emerging technologies in agriculture timeline

epochs. By comparison, after 40 training epochs, YOLO V8 had a mAP of 97.0%, an accuracy of 93.8%, and a recall of 96.0%. These results show that Roboflow outperformed YOLO V8 in the critical area of pest identification, which is essential for managing these agricultural risks.

Researchers [6] CNN implemented with Tensorflow to classify the Fall Armyworm. Due to data and resource constraints, they used Inception V3 pre-trained by transfer learning. The researchers collected field, lab, web, and augmented data and focused on categorising army worm during training. The model achieved an accuracy of 82%, with 32% false positives. The results suggest that automating pest identification using CNNs is a viable alternative to the labour-intensive manual method.

Researchers [7] introduced the 'Maize-Yolov5', an improved Yolov5 model, which is capable of automatically identifying budworms. Budworms are insects that can cause significant damage to maize fields. Early identification of this pest is crucial for implementing rapid control measures. The convolutional neural network model achieved a final accuracy of over 85%, indicating a positive outcome for monitoring pests in maize crops.

The authors [8] developed a convolutional neural network to classify different types of maize pests in photographs. They trained the neural network on the 'Maize Pests' dataset, which includes images of various maize pests. To achieve high levels of accuracy, it was crucial to train the network with a sufficient number of images. Additionally, they applied an image augmentation process to increase the amount of available data. During the training process, ten types of maize pests were categorised. In both training cases, an accuracy rate of over 90% was achieved.

III. METHODOLOGY

This chapter outlines the research process, including the main procedures and verification metrics used to ensure the reliability of the information. The research proposes an incremental methodology to solve the problem, with the aim of maintaining a constant search for high accuracy and completeness of the mAP. Quantitative data will be used for comparison and statistical analysis.

A. Summary of the Process

This chapter outlines the research methodology, primary measurements, and verification processes used to ensure data accuracy. To maintain a high level of correctness and completeness in the mAP, an incremental methodology is recommended. Quantitative data will be used for statistical analysis and comparison.. Figure 2 presents a summarized version of the process for the project realization.

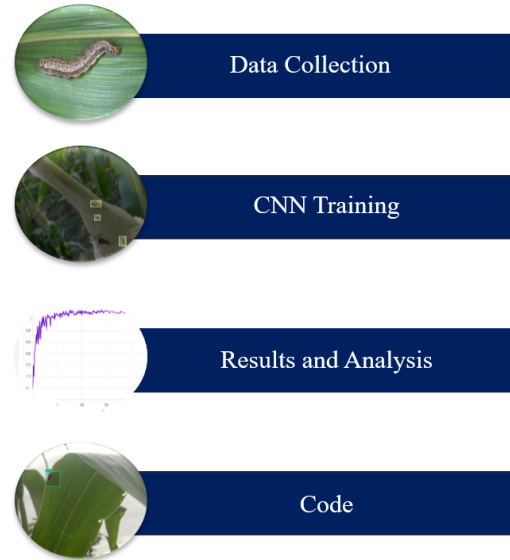


Fig. 2. Summary Process

1) *Collection and Preparation of Image Data sets:* As part of the research on maize pests, we will collect visual samples to create specific image sets. We have identified strategic resources, such as libraries and specialised data sets, to extract images that represent the different conditions that affect the presence of pests in maize. In the context of the modern economy, smart agriculture is a sophisticated stage of agricultural production [9]. Maize is a vital crop that is often at risk from various pests, which can significantly reduce both yield and quality. The primary objective of sample collection is to obtain images of the soil, maize plants, and pests identified in particular image libraries or data sets. Maize is the main staple food for the majority of the population in Honduras [10]. The model was specifically trained to recognize the three types of worms you mentioned (Fall Armyworm, Corn Stalk Borer, and Cutworms), but it is also capable of detecting Corn Flea Beetles, so the model can recognize at least four types of pests in total.

2) *Training Convolutional Neural Networks:* The development process is divided into increments and each phase is approached iteratively and cumulatively using the incremental methodology. This allows for continuous delivery of system improvements over time, distinguishing it from more traditional approaches. The model of the incremental methodology is presented in Figure 3.

This study considers 20 rounds of iterations. Changes are made to the classes, pre-processing, and data augmentation

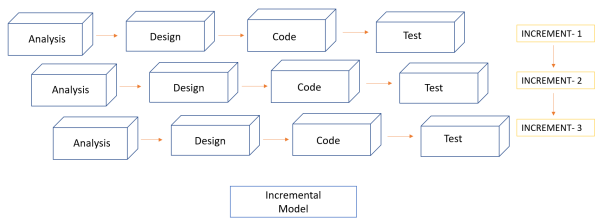


Fig. 3. Incremental Model

parameters during these iterations with the goal of achieving outstanding values in the evaluation metrics. It is important to note that Roboflow and YOLOv8 are consistently used throughout the process. The main challenge is to avoid obtaining metrics with low levels. Both models, YOLO V8 and Roboflow, were trained using periodic evaluation throughout the training process. This involved regular model evaluations using a validation dataset to measure key performance metrics such as precision, recall, and F1 score.

The final model selection was based on the performance observed during these periodic evaluations, as well as the overall performance on the training set throughout all training epochs.

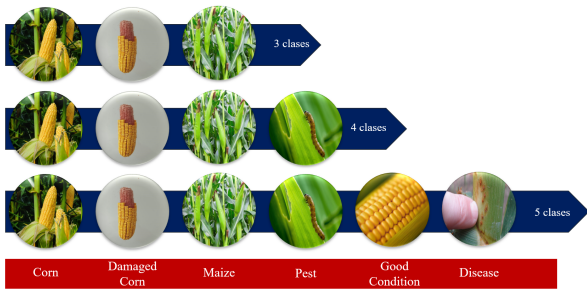


Fig. 4. Three, four, and six classes have been classified.

B. Validation Approach

Convolutional neural networks are trained over several epochs using a predetermined set of images. During the evaluation phase, important metrics such as accuracy, recall, and mean average precision are taken into account [11]. To ensure that neural networks can handle a wide range of scenarios, it is essential to include an adequate number of images. It is important to note that adding more features can help improve accuracy. Therefore, it is recommended to introduce variations not only in image quality and presentation but also in factors such as illumination and distance.

It is necessary to validate the collected samples to determine whether the features found in the maize match expectations. This involves exposing the convolutional neural network to a separate data set that is not used for training, allowing its ability to extrapolate patterns in new situations to be assessed.

IV. RESULTS

This section presents and discusses the data collected during the research. It includes information on the data set, its composition and characteristics, the quality and condition of the samples collected and tested, and the results of the proposed training. A detailed analysis of these aspects forms the basis of the results.

A. Dataset

The dataset initially comprises 1,227 USDA ARS images, as well as additional images of *Spodoptera frugiperda* and pest detection V2, resulting in a total of 4,773 images. To optimize inference and processing time during model training and facilitate the extraction of relevant patterns and features, the images were resized to 640x640 pixels. The partition assigns 80% of the images for training, 10% for validation during parameter tuning, and the remaining 10% for evaluating model performance on unseen data during the testing phase. This strategy aims to provide a comprehensive training set to ensure appropriate model fitting and evaluate model generalisability. Object detection models, such as YOLO, are designed to be robust against different aspect ratios of input images, including common ones such as 1:1, 16:9, or 4:3. By resizing the image to a specific size, such as 640 x 640 pixels, the objects retain their shape and relative proportions, regardless of the original aspect ratio of the image. This ensures that the model can accurately and consistently detect objects, regardless of variations in the aspect ratio of the input image.

B. CNN Trainings

The comparison of the 20 trainings is presented through graphs. The y-axis represents the evaluation metric, while the x-axis represents the number of rounds evaluated per class. Strategies such as non-maximum suppression (NMS), bounding box clustering, and prediction refinement were employed to address overlapping bounding boxes. Cross-validation was also performed to ensure model reliability and generalizability. The classes are divided into 3, 4, and 6. Figures 5, 6, 7, and 8 display the aforementioned graphs.



Fig. 5. Mean Average Precision (mAP) was calculated over 20 training sessions

The graph of mAP displays the percentages of the evaluated metrics, indicating that two out of three classes achieved a high level of individual accuracy. Therefore, the number of images entered into the CNN for classes three and four is sufficient.

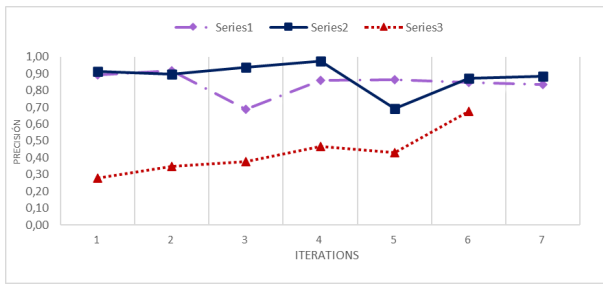


Fig. 6. Precision was calculated over 20 training sessions

The metric that produced the lowest results was precision; the percentages for classes 3, 4, and 6 show that two of the three classes attained a high degree of precision. The lowest level was recorded by the 6-class network, which had 8373 raw pictures in total.

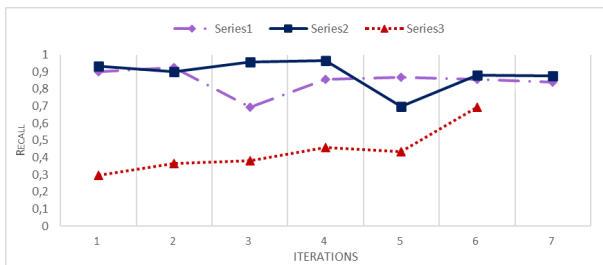


Fig. 7. Recall was calculated over 20 training sessions

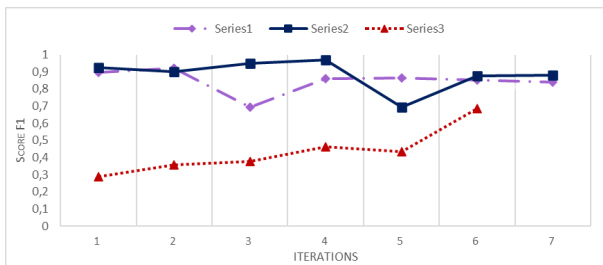


Fig. 8. Score F1 was calculated over 20 training sessions

The data analysis indicates that the neural network with four classes yielded the best results, likely due to a more balanced dataset. This suggests the potential for developing a more comprehensive model by incorporating a wider range of photos and accounting for distance and lighting differences. Out of the 4,773 photos entered, the network that produced the best results did not use augmentation. It showed the highest percentages in the assessment metrics, suggesting that these modifications might be the most appropriate for training neural networks using corn and pest data.

C. Lighting Conditions

The precision of object detection systems is significantly affected by illumination. The impact of different levels of illumination, measured in lux, on object detection accuracy

is shown in the table I. The data was obtained through tests conducted under controlled conditions, where illumination was varied and object detection accuracy was recorded.

TABLE I
IMPACT OF LIGHTING ON OBJECT DETECTION ACCURACY.

Lighting (Lux)	Accuracy (%)
24,000	0.790
9,000	0.900
1,000	0.890
500	0.920
12,000	0.875
3,000	0.940

The relationship between various levels of illumination and the accuracy of the pest detection model is illustrated in Table I. Various lighting conditions were examined, ranging from lower to very high levels, to observe how these conditions affect the model's ability to make accurate detections. The table demonstrates how variability in illumination can impact the model's performance. Figure 9 showcases pest detection under an illumination of 1000 lux.



Fig. 9. Impact of Lighting on Detection Accuracy.

V. CONCLUSION

The study's results demonstrate the effectiveness of the implemented convolutional neural network (CNN)-based model in identifying pests in maize crops. The model surpassed benchmark tests, achieving a mAP of 98.1%, an accuracy of 97.5%, and a recall of 96.8%. The CNN can facilitate further research on maize pests, thereby enhancing agricultural industry procedures. Among the pests studied, the whitefly (*Bemisia tabaci*) was found to be the most challenging to identify and train, indicating its complexity compared to other pests. However, the fall armyworm (*Spodoptera frugiperda*) was identified and trained most effectively among the pests. Accurate and early identification of pests is crucial for the agricultural sector to function optimally. This enables the quick application of control measures, leading to improved crop quality and yields. Computer vision and deep learning techniques provide farmers with a fast and non-intrusive method of collecting data.

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